

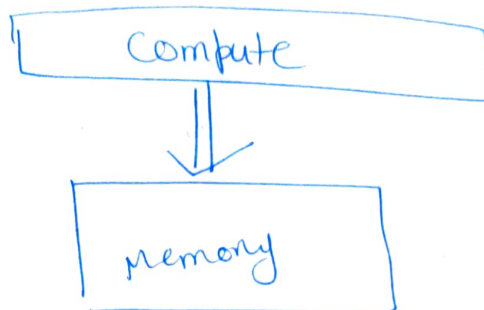
30-oct-25

GPU
SIMD
VLIW
pipelining

Parallel in nature
then only can
exploit any of
these features

Arithmetic / floating

MMM
MUM



In general communication is lot more expensive
than computation

on GPU can you do sort

SEARCH

Semineumerical stuff

→ (or) Sorting & Searching

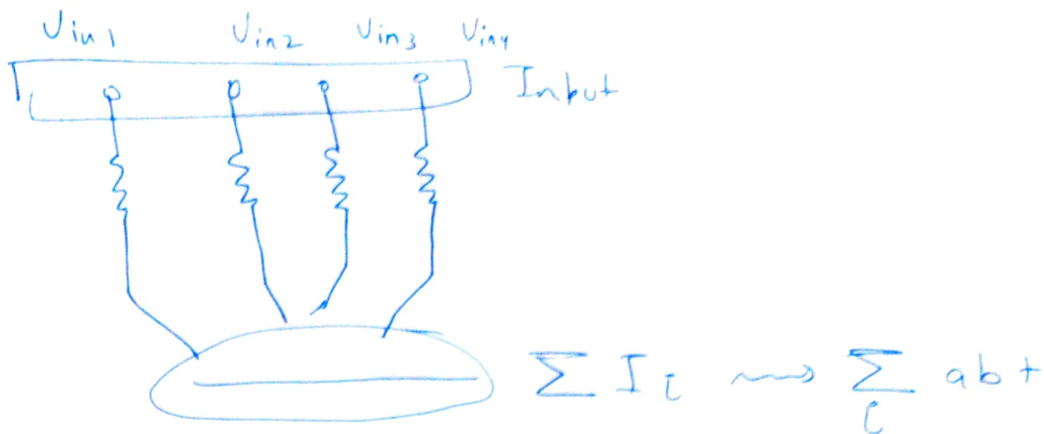
Can GPU be parallelised → yes

Search can be accelerated with GPU

GPU has two types of cores

└ CUDA → MAC

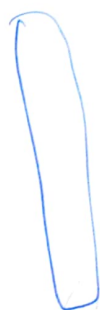
└ Tensor → $a \times b$ ABtc



Quantum is Analog computation

- very fast
- low —————

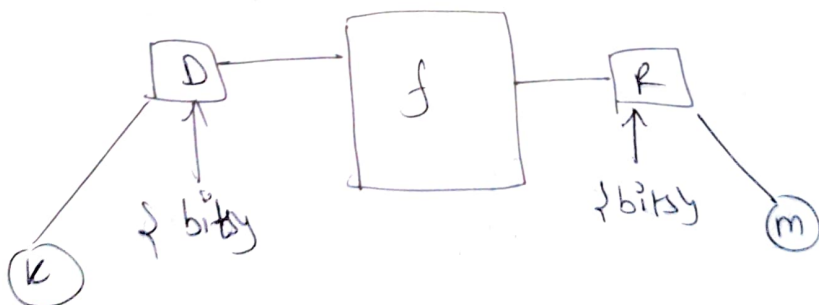
(i_1, i_2, i_3, i_4)



w_{11}	w_{12}	w_{13}	w_{14}
w_{21}	w_{22}	w_{23}	w_{24}
w_{31}	w_{32}	w_{33}	w_{34}
w_{41}	w_{42}	w_{43}	w_{44}

Domain specific

→ we want to use features of this and try and tune our Architecture



Don't really need
a floating point unit
 $m \geq k$ $k \neq m$

Convolution

- ↳ matrix / signal / EE
- ↳ matrix vector Multiplication

Accelerating SORT :

$O(n \log n) \rightsquigarrow$ comparison based sort
in general $O(n^2)$

$$O(n) \rightsquigarrow O(n)$$

for $i = 1 \text{ --- } N$

for $j = 1 \text{ --- } N$

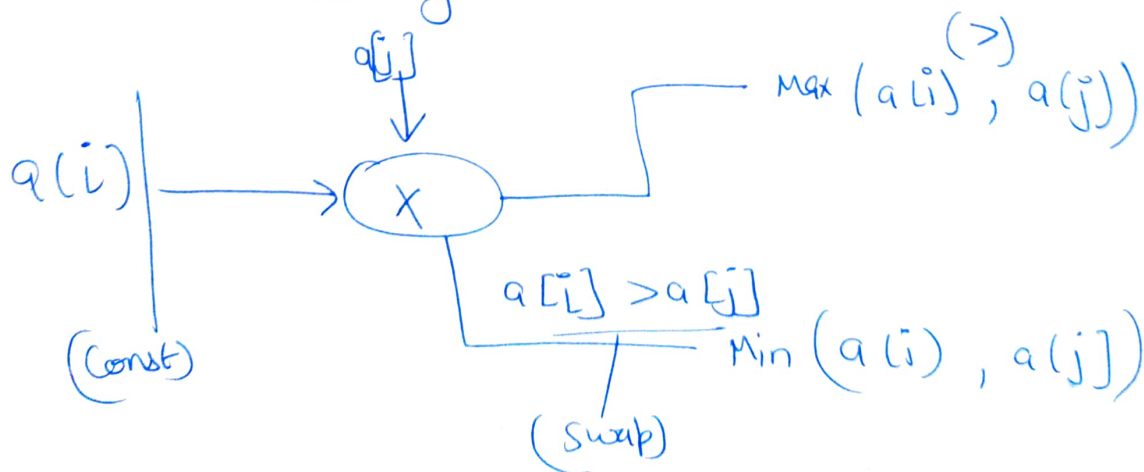
if $(a[i] > a[j])$

then

swap $(a[i], a[j])$

else

nothing



Systolic — ~~wave~~ wave, Does really come up in
Sorting Architecture (pic capped in side)

$$\rightarrow O(2n-1)$$



Data move moment b/w neighbours

$$\begin{bmatrix} a_{1j} \\ a_{2j} \\ a_{3j} \\ \vdots \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$O(n^2)$$

Given n processors $\rightarrow$ should be in $O(n)$

$$a_{11}x_1$$

$$a_{21}x_2$$

$$a_{31}x_3$$

$$a_{21}x_2 + y$$

$$a_{22}x_2 + y$$

$$a_{31}x_3 + y$$